Complete Robotics & High-Salary Engineering Learning Path

This document is a structured roadmap for Aryan Kumar Singh, a Mechanical Engineering student, to transition into a high-value Robotics and Autonomous Systems Engineer capable of targeting ■20–25+ LPA roles through strong technical skills, real-world projects, and deep-tech expertise.

1. The Main Goal

- Become a complete systems engineer, not only a mechanical engineer.
- Develop skills across robotics, embedded systems, AI, and automation.
- Focus heavily on practical implementation and project building.
- Target product companies, robotics startups, and deep-tech organizations.

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2. Exact Skill Priority Order

- Linux
- Python
- C++
- Git & GitHub
- Electronics basics
- ESP32 / STM32
- ROS2
- Control systems & PID
- OpenCV
- Embedded Systems
- SLAM & Navigation
- AI & Computer Vision

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3. First 3 Months Learning Path

- Learn Linux terminal usage and commands.
- Master Python basics deeply.
- Understand variables, loops, functions, OOP concepts.
- Start Git and GitHub.
- Learn C++ fundamentals.
- Build small ESP32 projects.
- Understand sensors and motors.

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4. Beginner Projects

- LED automation system
- Bluetooth controlled car
- Obstacle avoidance robot
- Servo motor controller
- Line follower robot
- Basic IoT dashboard

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5. Months 4–8 Learning Path

- Learn ROS2 architecture and communication.
- Understand nodes, topics, publishers, subscribers.
- Study PID control.
- Learn OpenCV for computer vision.
- Understand motor drivers and encoders.
- Study LiDAR and ultrasonic sensors.
- Learn basic path planning.

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6. Intermediate Projects

- Autonomous rover
- Maze-solving robot
- Object detection robot
- Camera tracking robot
- Self-balancing robot
- Mapping robot

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7. Months 8–15 Advanced Learning

- Learn SLAM deeply.
- Understand sensor fusion.
- Learn navigation stacks in ROS2.
- Study AI integration in robotics.
- Understand drone and autonomous systems.
- Contribute to open-source robotics projects.

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8. Advanced Projects

- Autonomous navigation robot
- Warehouse automation robot
- Computer vision inspection system
- AI-based robotic arm
- Drone navigation system
- Smart battery monitoring system

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9. Best Resources

- Python: Corey Schafer, freeCodeCamp
- C++: The Cherno
- Linux: NetworkChuck
- ROS2: Articulated Robotics
- OpenCV: Murtaza’s Workshop
- Embedded Systems: Phil’s Lab
- Git & GitHub: freeCodeCamp

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10. What Companies Actually Look For

- Strong projects
- Problem-solving ability
- Hands-on engineering skills
- GitHub portfolio
- Real hardware experience
- Ability to debug systems
- Consistency and initiative

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11. Internship Strategy

- Apply aggressively from 3rd year.
- Do not wait for campus placements.
- Build LinkedIn professionally.
- Maintain GitHub portfolio.
- Participate in hackathons.
- Contact professors and startups directly.

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12. How to Reach ■20–25+ LPA

- Develop rare interdisciplinary skills.

- Build high-quality robotics projects.
- Become strong in coding and embedded systems.
- Target product companies instead of service companies.
- Apply off-campus consistently.
- Prepare for technical interviews deeply.

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13. Important Reality

- This path is difficult but highly rewarding.
- Consistency matters more than motivation.
- Projects matter more than certificates.
- You will improve by building and debugging.
- Deep skills create high salaries.

Expected Career Growth Timeline

Year	Expected Progress
2nd Year	Strong fundamentals + beginner robotics projects
3rd Year	Advanced projects + internships + ROS2 expertise
Final Year	Portfolio-ready engineer with specialization
1–2 Years Job Experience	High-growth robotics/deep-tech engineer

Final Advice: Do not try to learn everything at once. Focus on consistent improvement every single week. Your goal is to become someone who can build intelligent engineering systems independently. That combination is rare — and rare engineers are paid extremely well.